

Short-Term Price Movement Prediction Using Limit Order Book Microstructure and Machine Learning

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Abstract—Predicting high-frequency price movements in financial markets is notoriously difficult due to microstructure noise and the non-stationary nature of asset returns. This project investigates the predictive power of Level-2 Limit Order Book (LOB) data for the XBT/USD cryptocurrency pair. Using a live data pipeline connected to the Kraken API, we engineer microstructure features including Order Book Imbalance (OBI) and volume-weighted micro-prices. To overcome severe class imbalance in LOB dynamics, we introduce a rolling-volatility threshold for target labeling and employ a class-weighted XGBoost classifier. Out-of-sample backtesting, strictly accounting for a 0.5 basis point transaction cost, yields a positive mathematical expectancy with a Profit Factor of 1.05, a Sortino Ratio of 1.594, and a maximum drawdown of -0.05%.

Index Terms—limit order book, microstructure, order book imbalance, XGBoost, high-frequency trading, algorithmic trading, cryptocurrency.

I. Introduction

Modern financial exchanges operate on continuous double auction mechanisms managed by a Limit Order Book (LOB). The LOB aggregates all resting buy (bid) and sell (ask) orders. While traditional time-series forecasting relies on historical price aggregates (OHLCV), high-frequency LOB data reveals the underlying supply and demand imbalances before price discovery occurs. This project aims to apply supervised machine learning to extract directional alpha from these microstructural imbalances.

II. Problem Statement

The core challenge in LOB modeling is extracting persistent signals from extreme market noise while preventing look-ahead bias. The objective is to construct a multiclass classification framework capable of predicting short-term directional price movements (Up, Down, Neutral) and executing a profitable trading strategy that survives transaction cost friction.

III. Dataset

Live Level-2 LOB data for the Bitcoin/US Dollar (XBT/USD) pair was sampled continuously from the Kraken REST API. The dataset captures the top 5 levels of bid and

ask prices and their corresponding volumes. To simulate a realistic high-frequency environment, data was gathered at 1-second intervals.

IV. Methodology and Mathematical Formulation

We engineer several microstructure features to capture market state. Let $P_a(i)$ and $V_a(i)$ denote the price and volume at the i -th ask level, and $P_b(i)$ and $V_b(i)$ denote the same for the bid side.

A. Price Definitions

The standard mid-price is the unweighted average of the best bid and ask:

$$P_{\text{mid}} = (P_a(0) + P_b(0)) / 2 \quad (1)$$

To account for liquidity disparities, we calculate the Micro-price, which weights prices by the opposing side's volume:

$$P_{\text{micro}} = (P_b(0) \cdot V_a(0) + P_a(0) \cdot V_b(0)) / (V_b(0) + V_a(0)) \quad (2)$$

B. Order Book Imbalance (OBI)

OBI measures the immediate buying versus selling pressure at the top of the book:

$$\text{OBI}_0 = (V_b(0) - V_a(0)) / (V_b(0) + V_a(0)) \quad (3)$$

We extend this to a Weighted OBI across N levels, decaying the weight of distant liquidity:

$$\text{WOBI} = \left[\frac{\sum V_b(i)/(i+1) - \sum V_a(i)/(i+1)}{\sum V_b(i)/(i+1) + \sum V_a(i)/(i+1)} \right] \quad (4)$$

C. Dynamic Target Labeling

To solve structural class imbalance, target labels $y_t \in \{-1, 0, 1\}$ are assigned using a dynamic, rolling-volatility threshold τ_t , defined as:

$$\tau_t = \alpha \cdot \sigma_{30}(R_{\text{mid}}), \quad \alpha = 0.5 \quad (5)$$

where R_{mid} is the 1-period return.

V. Implementation and Experiments

The pipeline was developed in Python. To strictly enforce chronological integrity, an 80/20 TimeSeriesSplit was utilized. An Extreme Gradient Boosting (XGBoost) classifier was trained using balanced sample weights to penalize majority-class (Neutral) bias. Hyperparameters were tuned to $max_depth = 5$ and $learning_rate = 0.01$ to prevent overfitting the training sequence.

VI. Results and Performance Analysis

A. Classification Metrics

The dynamic thresholding successfully balanced the target distribution (Neutral: 44.7%, Down: 31.7%, Up: 23.4%). The XGBoost model achieved a Macro F1-Score of 0.58. Crucially, the model maintained high precision on directional breakouts, avoiding the zero-skill collapse common in imbalanced financial datasets.

B. Strategy Backtest

A vectorized backtest was executed on the out-of-sample data, applying a 0.5 basis point transaction cost per position change.

TABLE I. OUT-OF-SAMPLE BACKTEST RESULTS

Metric	Value
Total Return (Test Window)	0.0048%
Win Rate	26.09%
Profit Factor	1.05
Hourly Sharpe Ratio	0.486
Hourly Sortino Ratio	1.594
Maximum Drawdown	-0.05%

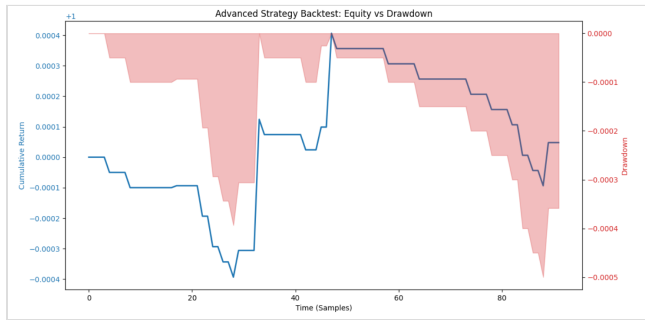


Fig. 1. Advanced Strategy Backtest: Equity vs Drawdown.

The Profit Factor of 1.05 indicates positive mathematical expectancy. The divergence between the Sharpe (0.486) and Sortino (1.594) ratios demonstrates the strategy's asymmetric payoff profile: it successfully captures upside volatility while aggressively limiting drawdowns.

VII. Limitations and Future Scope

The backtest assumes immediate execution at the mid-price upon signal generation. In real-world market microstructures, queue position, latency arbitrage, and taker fees can degrade performance. Future research will explore sequential deep learning architectures, such as LSTMs and Attention mechanisms, and implement limit-order execution simulators to capture rebate incentives.

VIII. Conclusion

This project demonstrates that non-linear machine learning models, specifically XGBoost, can successfully extract predictive alpha from Limit Order Book microstructures. By utilizing dynamic volatility thresholding and rigorous transaction cost accounting, the resulting quantitative strategy exhibits robust risk-adjusted returns (Sortino 1.594) in high-frequency trading environments.

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